

TODP V1 Architecture Design Document (ADD)

Version: 1.0

Status: Frozen

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1. Architecture Overview

Purpose

This document defines how TODP V1 is implemented.

The PRD defines:

“What the system must do.”

The ADD defines:

“How the system works.”

Architectural Goals

The architecture must support:

- Traceability
- Auditability
- Provenance
- Explainability
- Reproducibility
- Evidence-Based Recommendations

Architectural Constraints

The system must not:

- Invent sources
- Invent citations
- Invent metrics
- Invent authority structures

Every recommendation must be traceable to evidence.

2. High-Level System Architecture

User



Frontend



API Gateway



Orchestration Layer



Agent Layer



Retrieval Layer



Evidence Layer



Storage Layer

Core Layers

Presentation Layer

Responsible for:

- Search
 - Discovery
 - Recommendation Display
 - Evidence Display
 - Audit Display
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Orchestration Layer

Responsible for:

- Workflow Coordination
 - Agent Invocation
 - Result Aggregation
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Agent Layer

Responsible for:

- Discovery
 - Ranking
 - Provenance
 - Audit
 - Citation Generation
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Storage Layer

Responsible for:

- Opportunities
 - Evidence
 - Sources
 - Audit Logs
 - Recommendations
-
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3. Agent Architecture

Design Principle

Every major responsibility belongs to a dedicated agent.

No single agent owns the entire workflow.

Agent List

Discovery Agent

Responsible for:

- Opportunity Discovery
 - Event Discovery
 - Scholarship Discovery
 - Course Discovery
 - Government Program Discovery
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Authority Discovery Agent

Responsible for:

- Discover Authority Structures
- Discover Jurisdiction Hierarchies
- Validate Authority Relationships
- Update Authority Structures

Examples:

Institution

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Local

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District

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State

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National

↓

International

Provenance Agent

Responsible for:

- Source Tracking
 - Evidence Tracking
 - Metadata Collection
 - Retrieval Tracking
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Ranking Agent

Responsible for:

- Money Cost Calculation
 - Time Cost Calculation
 - Distance Cost Calculation
 - Ranking Score Calculation
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Citation Agent

Responsible for:

- Citation Generation
 - Citation Validation
 - Source Linking
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Registration Agent

Responsible for:

- Form Detection
- Auto Fill
- User Confirmation

Never Auto Submit.

Calendar Agent

Responsible for:

- Calendar Sync
- Availability Checks
- Event Scheduling

Requires User Consent.

Hallucination Audit Agent

Responsible for:

- Missing Source Detection
 - Missing Citation Detection
 - Unsupported Claim Detection
 - Confidence Verification
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Recommendation Audit Agent

Responsible for:

- Recommendation Verification
 - Ranking Verification
 - Provenance Verification
 - Audit Report Generation
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4. Authority Hierarchy Architecture

Purpose

Authority structures must never be hardcoded.

Authority structures must be discovered, stored and updated.

Authority Discovery Workflow

Authority Source

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Authority Discovery Agent

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Hierarchy Validation

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Authority Store

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Versioning



Audit

Supported Authority Types

- Government
 - Educational Institutions
 - NGOs
 - Companies
 - International Organizations
 - Public Bodies
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Example Government Hierarchy

Municipality



District



State Department



State Government



Central Ministry



Government of India

Example Education Hierarchy

College



University



Regulator



Ministry

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5. Universal Recommendation Pipeline

Every recommendation follows:

Discovery



Evidence Retrieval



Source Retrieval



Citation Retrieval



Timestamp Retrieval



Money Cost Calculation



Time Cost Calculation



Distance Cost Calculation



Ranking



Audit



Display

Ranking Formula

Total Cost Score =

Money Cost

- Time Cost
- Distance Cost

Online Opportunity Rule

Online opportunities participate in the same ranking system.

For online opportunities:

Distance Cost = 0

Travel Time = 0

Universal Recommendation Metadata

Every recommendation must contain:

- Recommendation ID
- Source ID
- Citation ID
- Evidence Chain ID
- Retrieval Timestamp
- Calculation Method
- Audit Status
- Version
- Money Cost
- Time Cost
- Distance Cost
- Ranking Score

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END OF ADD V1 SECTIONS 1–5